

2026 ISPE ANNUAL MEETING

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Let's Get Personal:

Defining and estimating effects
of dynamic treatment strategies
using real-world data

42nd Annual Meeting

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Lucia Petito, *Northwestern University*



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INTERNATIONAL SOCIETY FOR PHARMACOEPIDEMIOLOGY
PRE-CONFERENCE COURSE 2026

Let's Get Personal

Defining and Estimating Effects of Dynamic Treatment Strategies Using Real-World Data

Emma McGee, Lucia Petito, Rienna Russo

August 30, 2026

2:30 PM – 6:00 PM

Teaching Team



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INTERNATIONAL SOCIETY FOR PHARMACOEPIDEMIOLOGY
PRE-CONFERENCE COURSE 2026

Let's Get Personal

Defining and Estimating Effects of Dynamic Treatment Strategies Using Real-World Data

Session 1

Rienna Russo, Emma McGee, Lucia Petito

Outline

- **Motivation**

Role in personalized medicine

- **Defining dynamic treatment strategies**

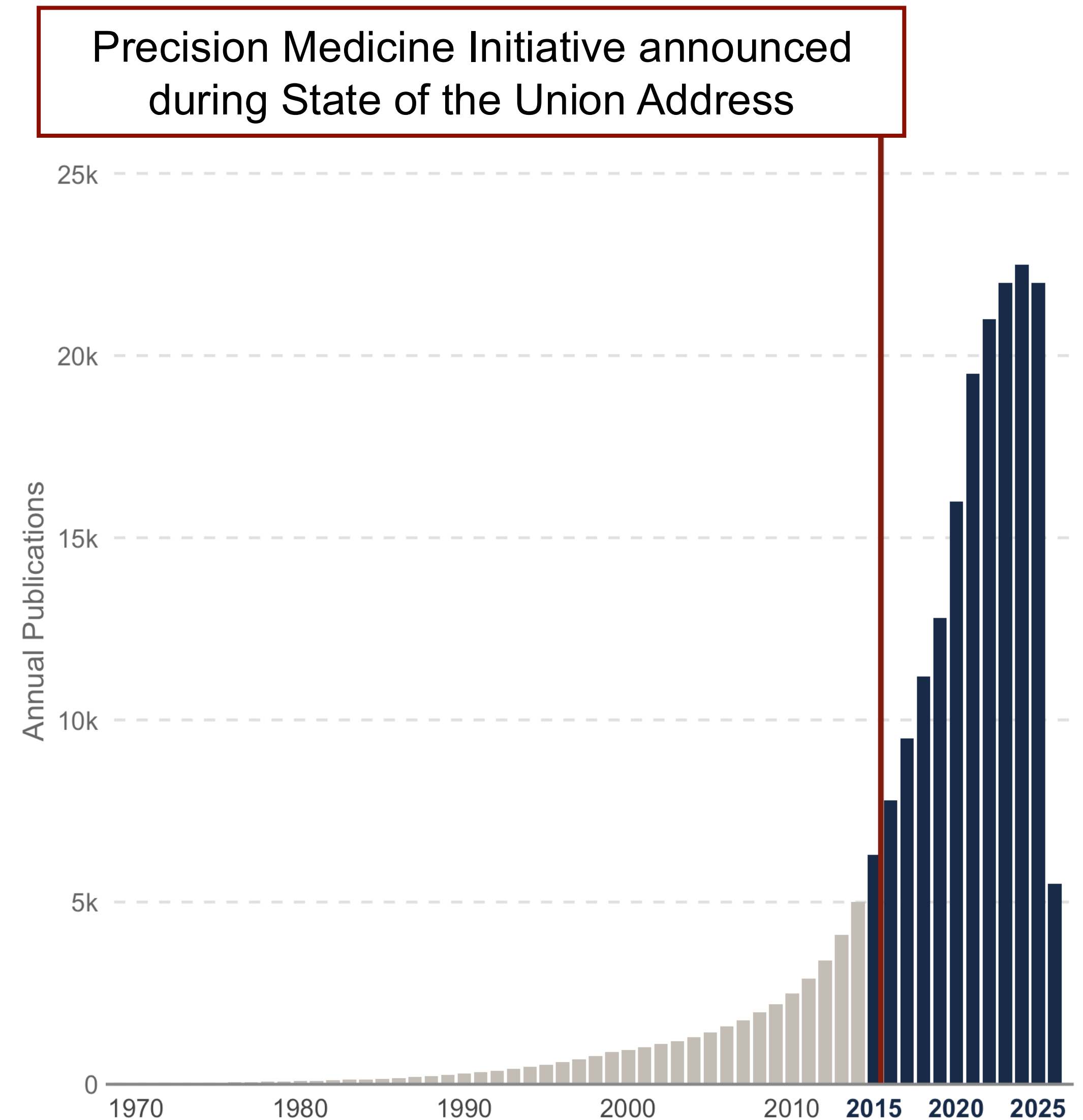
Eligibility criteria and treatment strategies

Common pitfalls and utility of the target trial framework

Distinguishing estimands

Interest in Precision Medicine

Source: PubMed
Search term: "precision medicine"
* 2026 partial data through April 2026



One size does not fit all

Target the right treatments to the
right patients at the **right time**.

Right patients

Heterogeneous treatment effects

Right time

Dynamic treatment strategies

One size does not fit all

For people with diabetes and hypertension, blood pressure goals should be made through shared decision-making process that addresses cardiovascular risk, potential adverse effects of antihypertensive medications, and individual preferences.

Standards of Care in Diabetes (2026) American Diabetes Association

The optimal duration of neoadjuvant endocrine therapy ... should be individualized and guided by careful evaluation of the patient's clinical status and the clinical response over time.

Neoadjuvant Chemotherapy, Endocrine Therapy, and Targeted Therapy for Breast Cancer Guideline (2021) American Society of Clinical Oncology

Individualize initial treatment approaches for obesity (i.e., lifestyle and nutritional therapy, pharmacologic agents, or metabolic surgery) based on the person's medical history, life circumstances, preferences, and motivation.

Standards of Care in Overweight and Obesity (2026) American Obesity Association

Treatment vs. Treatment Strategy

Treatment may refer to the medication of interest, which could be just one component of an overall treatment strategy.

A **treatment strategy** refers to the set of decision rules that specify the treatment assigned throughout follow-up.

- Treatment *policies*
- Treatment *regimes*

Static vs. Dynamic Strategies

Static strategies: which treatment patients will take at each point during follow-up is known when the treatment strategy is initially assigned at time zero and does not change after baseline.

Examples: Always treat, treat for one-year, never treat

Static vs. Dynamic Strategies

Static strategies: which treatment patients will take at each point during follow-up is known when the treatment strategy is initially assigned at time zero and does not change after baseline.

Examples: Always treat, treat for one-year, never treat

Dynamic strategies: which treatment patients will take at each point during follow-up will depend on their history up until that point, i.e., are based on evolving characteristics.

Scenarios that call for dynamic strategies

When to start

When to escalate

Head-to-head comparators

Inactive comparators

Scenarios that call for dynamic strategies

When to start

When to escalate

Head-to-head comparators

Inactive comparators

Comparative effectiveness of immediate antiretroviral therapy versus CD4-based initiation in HIV-positive individuals in high-income countries: observational cohort study

Sara Lodi, Andrew Phillips, Roger Logan, Ashley Olson, Dominique Costagliola, Sophie Abgrall, Ard van Sighem, Peter Reiss, José M Miró,

Treatment Strategies

- (1) Immediate initiation of ART
- (2) Initiation at CD4 count less than 500 cells per μL
- (3) Initiation at CD4 count less than 350 cells per μL

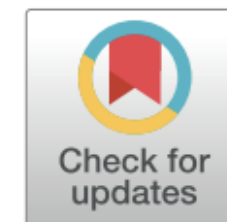
Scenarios that call for dynamic strategies

When to start

When to escalate

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Inactive comparators



Journal of Clinical Epidemiology 151 (2022) 96–103

**Journal of
Clinical
Epidemiology**

ORIGINAL ARTICLE

Thromboprophylaxis with standard-dose vs. flexible-dose heparin for hospitalized COVID-19 patients: a target trial emulation

Gonzalo Martínez-Alés^{a,*}, Arce Domingo-Relloso^{b,c}, Manuel Quintana-Díaz^d,
Carmen Fernández-Capitán^d, Miguel A. Hernán^{a,e}, COVID@HULP Group

Treatment Strategies

- (1) Standard dose: LMWH at standard prophylactic dose
- (2) Variable dose: Same, except that dose is increased within 48 hours if analytic abnormalities are detected.

Under both strategies, the choice of drug is left to the physician's discretion. Individuals are excused from following the assigned strategy after thrombosis (when anticoagulation treatment must be started) and after active bleeding or platelet count $<50,000/\text{mL}$ (when anticoagulation must be discontinued).

Scenarios that call for dynamic strategies

When to start

When to escalate

Head-to-head comparators

Inactive comparators

nature mental health

Article

<https://doi.org/10.1038/s44220-025-00407-5>

Comparative effectiveness of long-acting injectable versus oral antipsychotic medication after a first episode of psychosis

Treatment Strategies

- (1) Initiation of LAI aripiprazole or paliperidone
- (2) Continuation of oral aripiprazole or paliperidone.

Under both strategies, all treatment decisions are left at the physician's discretion after serious side effects, defined as tremor, akathisia, severe anticholinergic effects or an Abnormal Involuntary Movement Scale (AIMS) score or a Glasgow Antipsychotic Side Effects Scale (GASS) score greater than the site-specific 75th percentile

Scenarios that call for dynamic strategies

When to start

When to escalate

Head-to-head comparators

Inactive comparators



JNCI: Journal of the National Cancer Institute, 2026, 00(00), 1–9

<https://doi.org/10.1093/jnci/djag158>

Advance Access Publication Date: May 29, 2026

Article

Statins and risk of cardiovascular disease: emulating a primary prevention trial in breast cancer survivors

Rienna G. Russo , PhD, MHS, ScM^{1,2,3,*}, Miguel A. Hernán, MD, DrPH^{2,3,4}, Issa J. Dahabreh, MD, ScD^{2,3,4,5}, Eric B. Rimm, ScD^{3,6,7},

Treatment Strategies

- (1) No use of statins throughout follow-up, allowing initiation when low-density lipoprotein >160 mg/dL
- (2) Initiation of statins and continuation throughout follow-up, allowing switching to other statins or lipid-lowering treatments and discontinuation if clinically indicated
 - a) Creatinine $\geq$ 2 mg/dL
 - b) Albumin $\leq$ 3.5 mg/dL
 - c) ALT >36 U/L

Motivating Example

Glucagon-Like Peptide-1 Receptor Agonists (GLP-1RA)

- Class of medications used to treat type 2 diabetes (T2D) and obesity
 - Semaglutide (Ozempic for T2D, Wegovy for obesity)
 - Tirzepatide (Mounjaro for T2D, Zepbound for obesity)
- Contraindications to treatment include
 - Pancreatitis, pregnancy, severe gastrointestinal diseases (gastroparesis, inflammatory bowel disease), medullary thyroid cancer
- Growing interest in repurposing these medications for prevention and treatment of other conditions
 - Dementia, cancer, polycystic ovarian syndrome, metabolic dysfunction-associated steatohepatitis, etc.

Scenarios that call for dynamic strategies

When to start

Initiate GLP1-RA within a certain time after first elevated HbA1c measurement.

When to escalate

Add a GLP1-RA within a certain time after initiating metformin.

Head-to-head comparators

Initiate and continue GLP1-RA unless pancreatitis or medullary thyroid cancer develops vs. initiate and continue DPP4i unless renal impairment.

Inactive comparators

Initiate and continue GLP1-RA unless pancreatitis or medullary thyroid cancer develops vs do not initiate GLP1-RA and continue without unless $\text{HbA1c} \geq 7.5\%$.

Outline

- **Motivation**

Role in personalized medicine

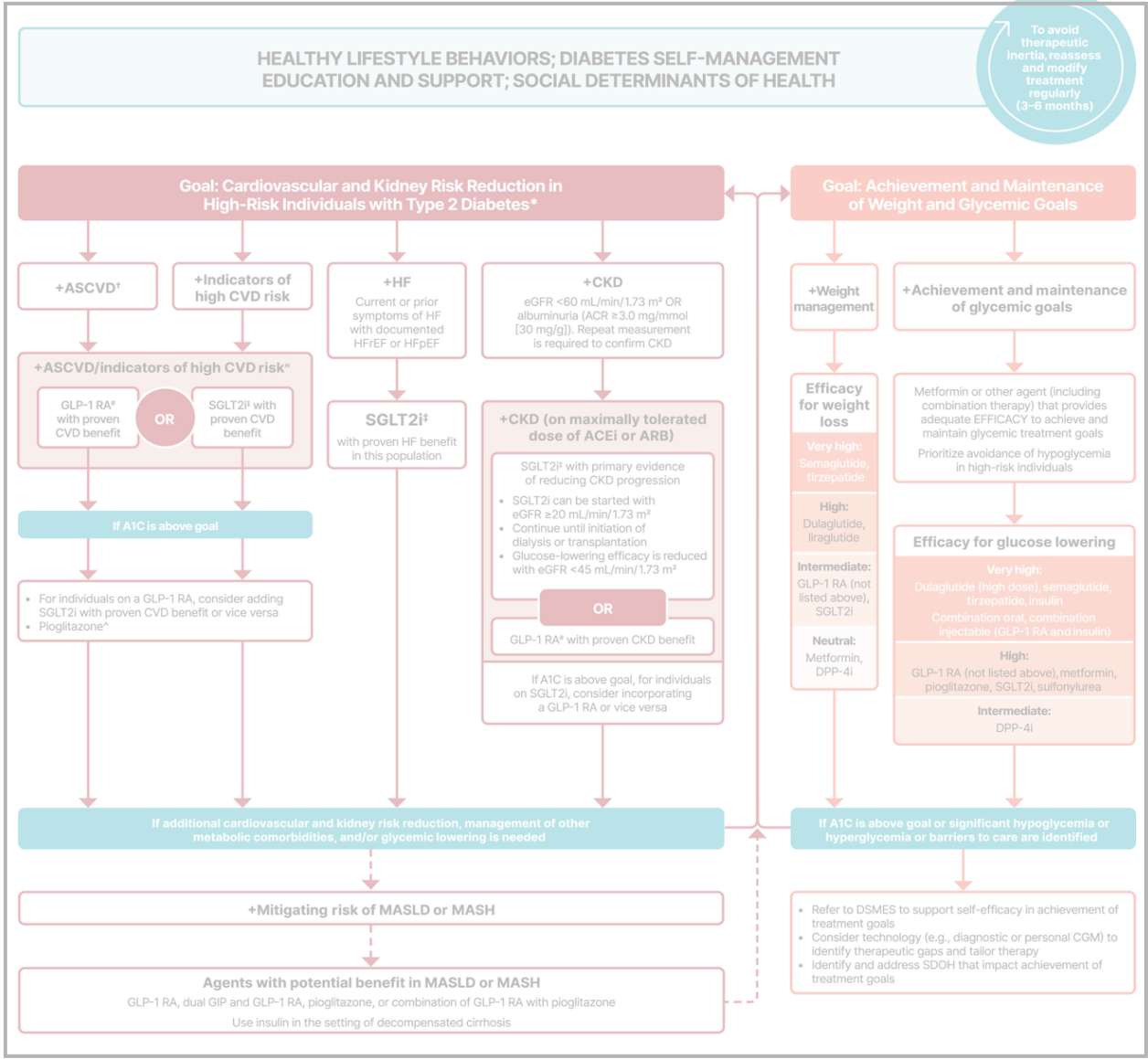
- **Defining dynamic treatment strategies**

Eligibility criteria and treatment strategies

Common pitfalls and utility of the target trial framework

Distinguishing estimands

Learning how treatment decisions are made



Clinical Guidelines

Standards of care and scientific statements from nonprofit professional and advocacy organizations outline clinical practice medications.

The drug label for OZEMPIC semaglutide injection provides detailed prescribing information. It includes highlights of prescribing information, warnings, contraindications, and precautions. The label also details dosage forms and strengths, recent major changes, indications and usage, dosage and administration, adverse reactions, drug interactions, and use in specific populations. The label is designed to be clear and easy to read, with a focus on patient safety and effective use of the medication.

Drug Labels

FDA drug labels include a wealth of information including *indications* and *contraindications*, *dosage* and *administration*, as well as adverse reactions and black box warnings.



Expert Knowledge

Clinicians with real-world experience treating patients can provide additional insights.



Eligibility criteria

Determine the target population to whom findings will be applied.

1. Identify clinically-relevant population

E.g., failed on first-line treatment

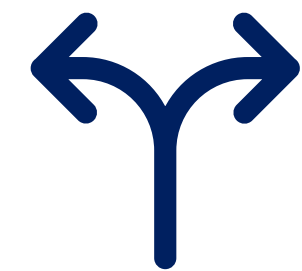
2. Restrict based on important covariates

E.g., indications (elevated A1c) and contraindications (pancreatitis)

Treatment Strategies

Explicitly state what is allowed and when for each alternative, e.g., start and stop times, duration, frequency, dosing.

1. Make reasonable comparisons to maintain clinical equipoise.
Similar lines of treatments, active comparators, standard of care.
2. Define treatment modifications allowed within the protocol.
Allow individuals to stop treatment when contraindicated, or start when indicated.



Example

Dual therapy Addition of GLP1 to SGLT2 with continued use of both throughout follow-up unless change is indicated.

VS

Monotherapy Continued use of SGLT2 without the addition of GLP1 at any time during follow-up unless change is indicated.

Under all treatment strategies:

- Discontinuation of treatment prior to contraindication is not allowed.
- Any study medication from the class assigned at any dose is allowed at any time (switch between or add medications).
- Treatment is considered continuous if there is a gap of < 30 days between successive prescriptions.
- At the discretion of the physician and patient,
 - Metformin can be continued or discontinued.
 - Switching to insulin monotherapy is allowed.
 - Medication class can be changed if severe urinary tract infection, diabetic ketoacidosis, pancreatitis, or medullary thyroid cancer develops.

Deviation

Allowable gap

Excused

Dynamic Treatment Strategies

Deviation

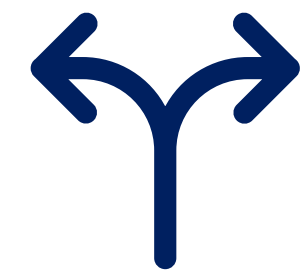
A **deviation** occurs when an individual's observed data are no longer compatible with the treatment strategy.

This can occur when an individual changes treatment without a clinical reason to do so.

E.g., Initiating treatment without an indication or discontinuing treatment without a contraindication.

Depending on the estimation procedure, individuals may be censored when they deviate from the assigned strategy.

E.g., Inverse probability weighting - more on this next lecture!



Dynamic Treatment Strategies

Allowable gap

When interested in sustained medication use, the **allowable gap** can bridge between successive prescriptions.

I.e., Time allowance between the end of one prescription coverage period and start of another to be considered continuous.

What is considered a deviation will depend on the allowable gap.

This is often a parameter that we vary to understand the sensitivity of estimates to changes in potential non-adherent periods, e.g., use 60 days in the main analysis, then 30 and 90 days in sensitivity analyses.

Dynamic Treatment Strategies

Excused

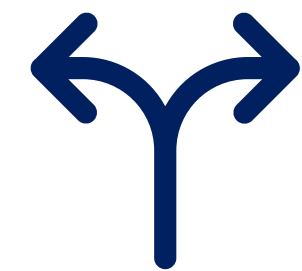
When an individual is **excused**, treatment decisions are up to the discretion of the clinician and patient.

Individuals follow the natural course, i.e., whatever they would otherwise have done.

Excused = no intervention.

This can occur after an individual develops an indication or experiences tolerability issues.

E.g., after a patient assigned to take SGLT2 inhibitor initiates renal replacement therapy, they may change their medication class at any time.



Example

Dual therapy Addition of GLP1 to SGLT2 with continued use of both throughout follow-up unless change is indicated.

VS

Monotherapy Continued use of SGLT2 without the addition of GLP1 at any time during follow-up unless change is indicated.

Under all treatment strategies:

- Discontinuation of treatment prior to contraindication is not allowed.
- Any study medication from the class assigned at any dose is allowed at any time (switch between or add medications).
- Treatment is considered continuous if there is a gap of < 30 days between successive prescriptions.
- At the discretion of the physician and patient,
 - Metformin can be continued or discontinued.
 - Switching to insulin monotherapy is allowed.
- At the development of severe urinary tract infection or diabetic ketoacidosis, SGLT2 should be discontinued, and at the development of pancreatitis, or medullary thyroid cancer, GLP1 should be discontinued.

Deviation

Allowable gap

Excused

Deviation

Outline

- **Motivation**

Role in personalized medicine

- **Defining dynamic treatment strategies**

Eligibility criteria and treatment strategies

Common pitfalls and utility of the target trial framework

Distinguishing estimands

Key considerations



Eligibility criteria

Identify clinically-relevant population.

- E.g., failed on first-line treatment

Make restrictions based on important covariates.

- E.g., indications (elevated A1c) and contraindications (pancreatitis)



Treatment strategies

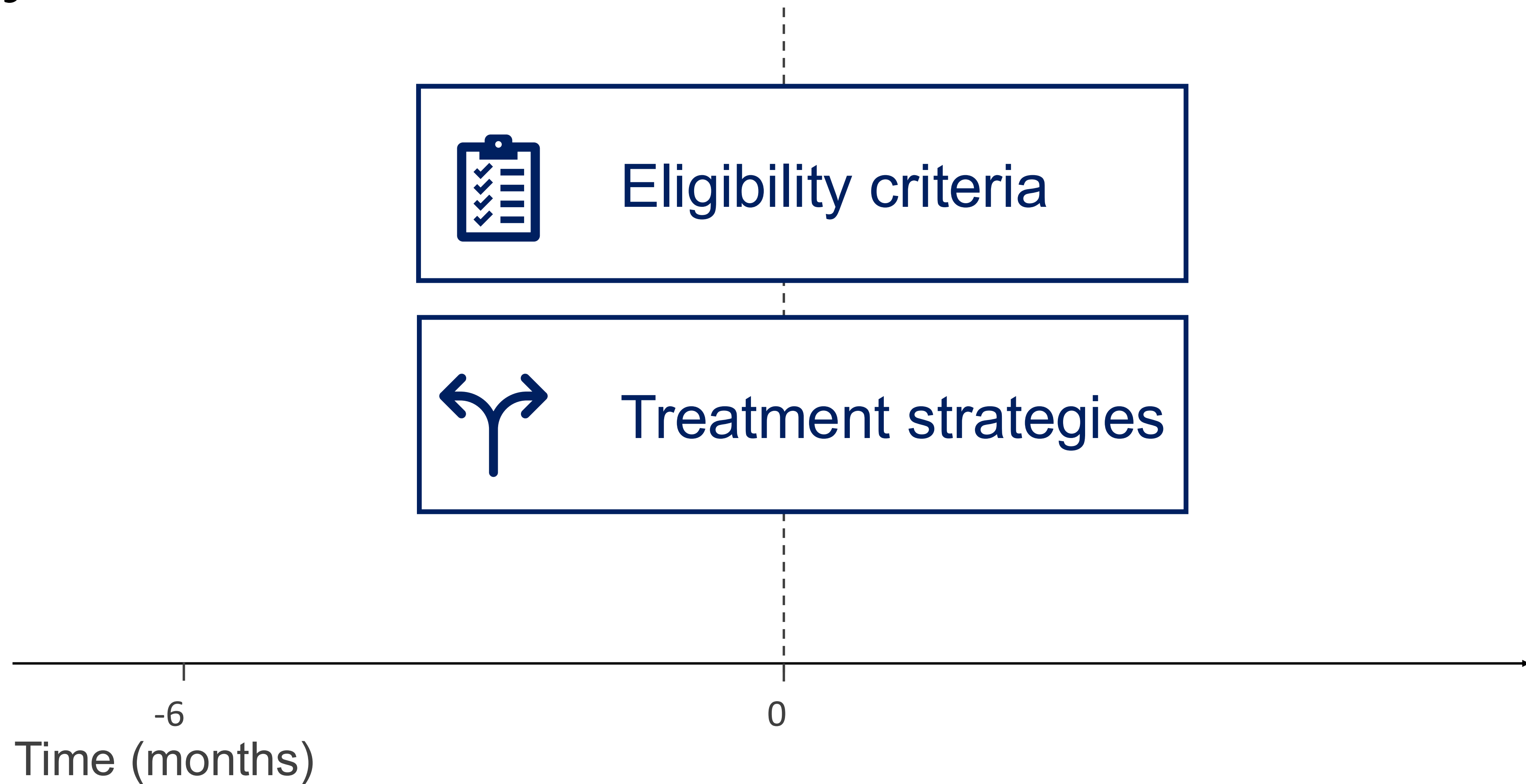
Make reasonable comparisons.

- E.g., same lines of treatments, active comparators, standard of care

Define treatment modifications allowed within the protocol.

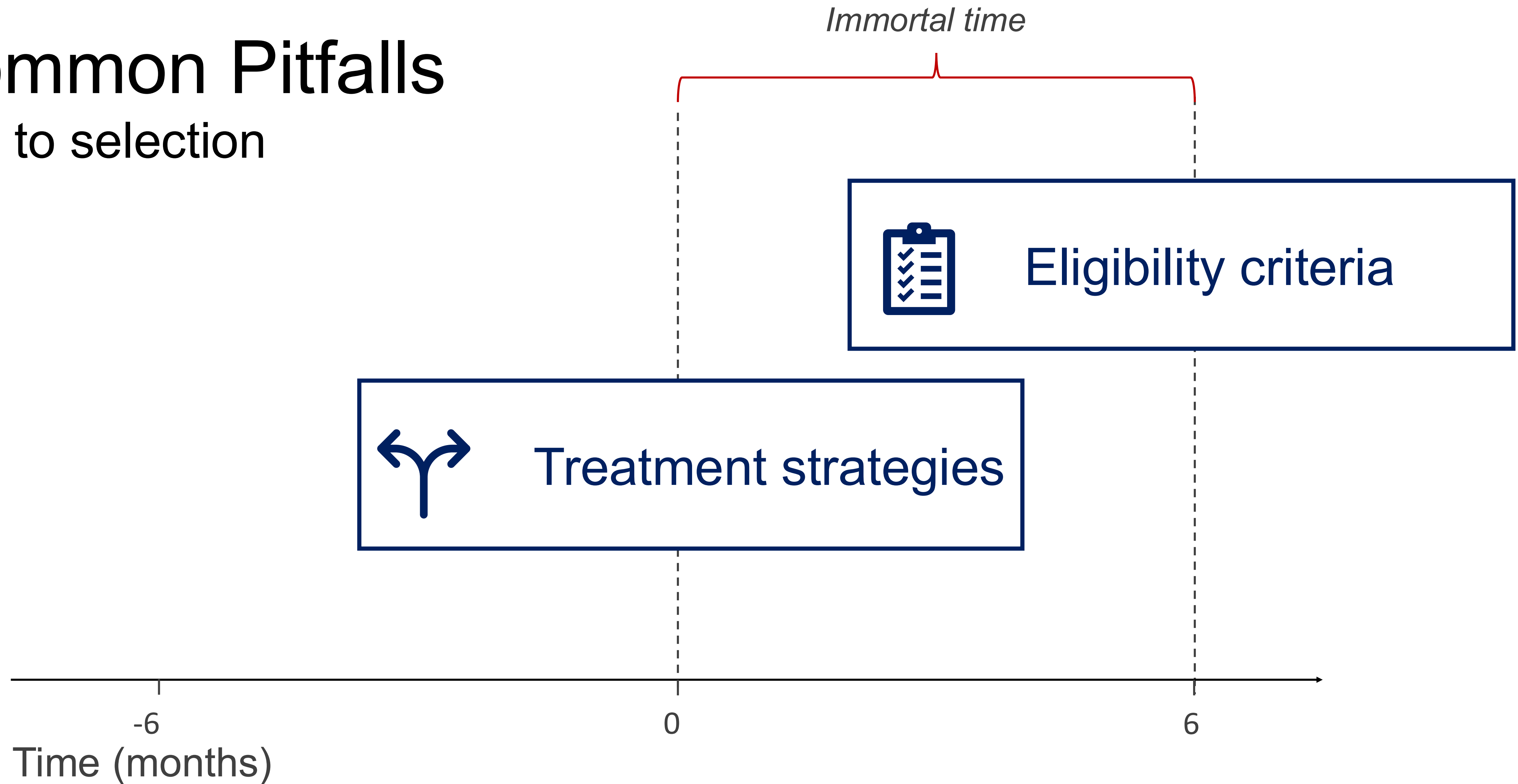
- E.g., Individuals who develop contraindications are unlikely to adhere

Key Considerations



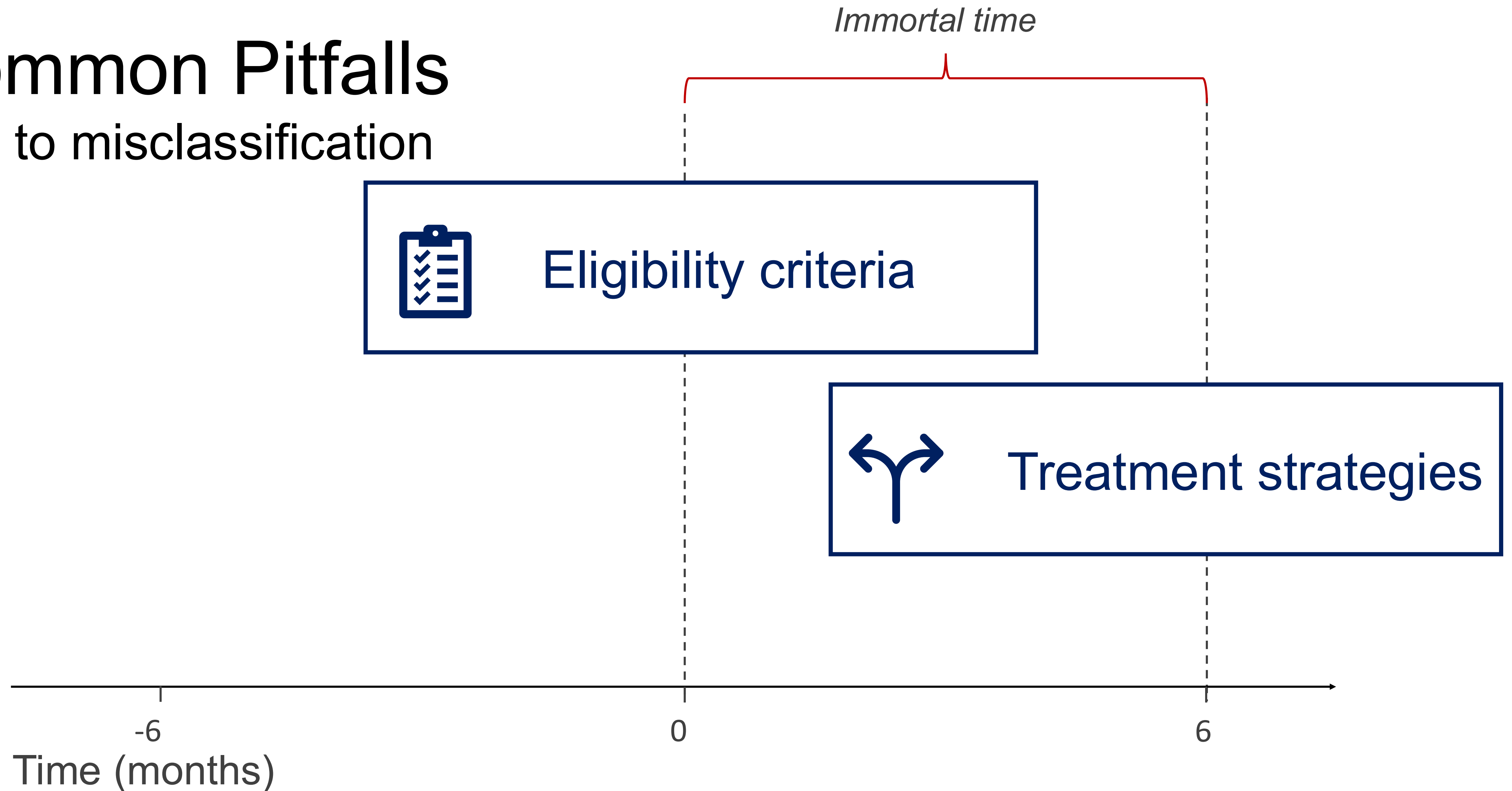
Common Pitfalls

Due to selection



Common Pitfalls

Due to misclassification



Common Pitfalls

Immortal time can arise when

- Eligibility criteria are applied after the start of follow-up
- Treatment strategies are assigned after the start of follow-up

These situations are more prone to occur when individuals meet the eligibility criteria at multiple times, since the time zero is not unique, and when treatment strategies are indistinguishable at baseline, since the assignment to treatment strategies is not unique.

Target Trial Emulation

- ① Ask a causal question
Specify the protocol



Target Trial Emulation

- ① Ask a causal question
Specify the protocol
- ② Answer the causal question
A. Conduct the trial
B. Use observational data
to **explicitly** emulate it



Revisiting Common Pitfalls

Aligning eligibility criteria, treatment assignment, and start of follow-up can be more challenging when

- Eligibility criteria met at multiple times
May make selecting time zero, and ensuring no eligibility criteria are applied after, harder
- Treatment strategies are indistinguishable at baseline
May make assigning treatment strategies without looking into the future harder

Eligibility criteria met at multiple times

There are two valid options for selecting time zero

1. A single eligible time (e.g., first or random)
2. All eligible times may be used

Option #2 is more statistically efficient, particularly when one treatment strategy is less common.

See Hernán et al. J Clin Epidemiol 2016 for more details on *sequential trial emulation*.

Treatment strategies indistinguishable at t_0

There are two valid options for selecting time zero

1. Assign a random strategy
2. All strategies may be assigned

Option #2 is more statistically efficient.

See Hernán BMJ 2018 for more details on *clone-censor-weighting*.

Outline

- **Motivation**

Role in personalized medicine

- **Defining dynamic treatment strategies**

Eligibility criteria and treatment strategies

Common pitfalls and utility of the target trial framework

Distinguishing estimands

Defining estimands

The per-protocol effect of a treatment does not exist – there are many different possible causal effects.

For a binary treatment that we would like to intervene on at k timepoints, there are at least 2^k *static* sustained strategies we could consider.

In a trial, a per-protocol effect is defined as the effect of adhering to the assigned treatment strategies as specified in the protocol.

Defining estimands

Target trial emulation involves clearly specifying the protocol for what is and is not allowed under each treatment strategy when using real-world data to emulate the hypothetical trial.

Non-naïve per-protocol analyses of randomized trials in the presence of non-adherence are identical to per-protocol analyses of observational studies.

With the caveat of baseline confounding in observational analyses.

Defining estimands

When using real-world data, investigators **assign** patients to treatment strategies based on data compatibility at baseline and **adherence** to the treatment strategies based on data compatibility over follow-up.

→ *Observational analogue of the per-protocol effect*

Assignment and **adherence** may be operationalized using the same data source:

1. Medical records, i.e., what is prescribed by the clinician.
2. Dispensing records, i.e., what is filled in the pharmacy.
3. Pharmacy claims, i.e., what is covered by insurance

A note on other common contrasts

As-treated classifies patients according to the treatment that they took rather than the treatment they were assigned to take.

A contrast of those who took treatment and those who did not, regardless of their assigned treatment strategy.

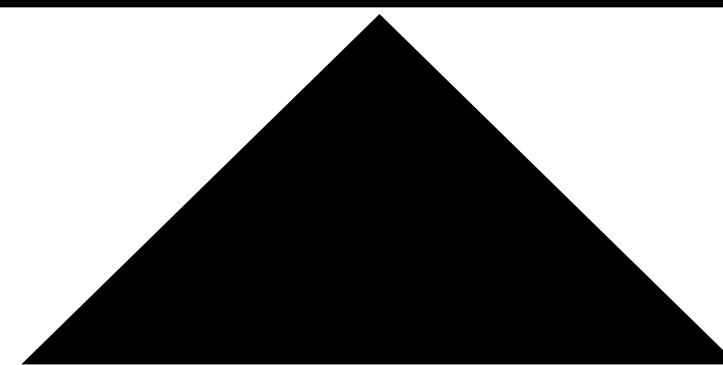
On-treatment only includes data from patients when they are actively taking the study treatment to which they were assigned.

Differs from 'per-protocol' in that the contrast is based on adherence to the assigned treatment rather than *strategy*, i.e., fails to account for clinical circumstances that may necessitate medication change.

The trade-off

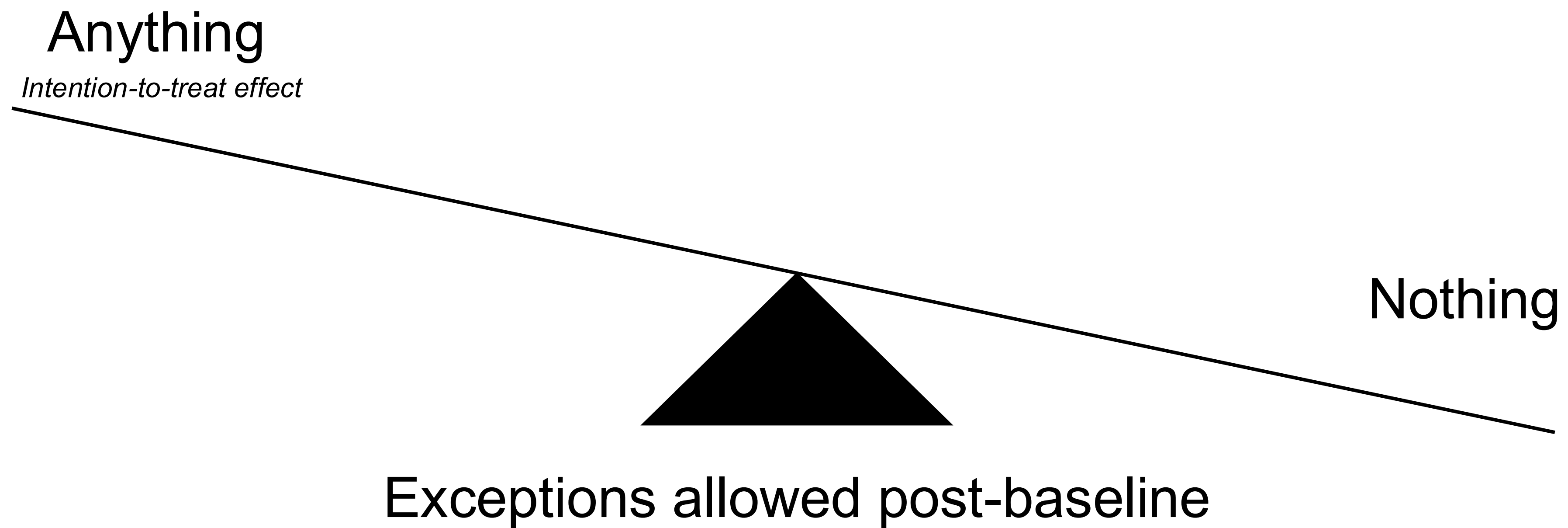
Anything

Nothing

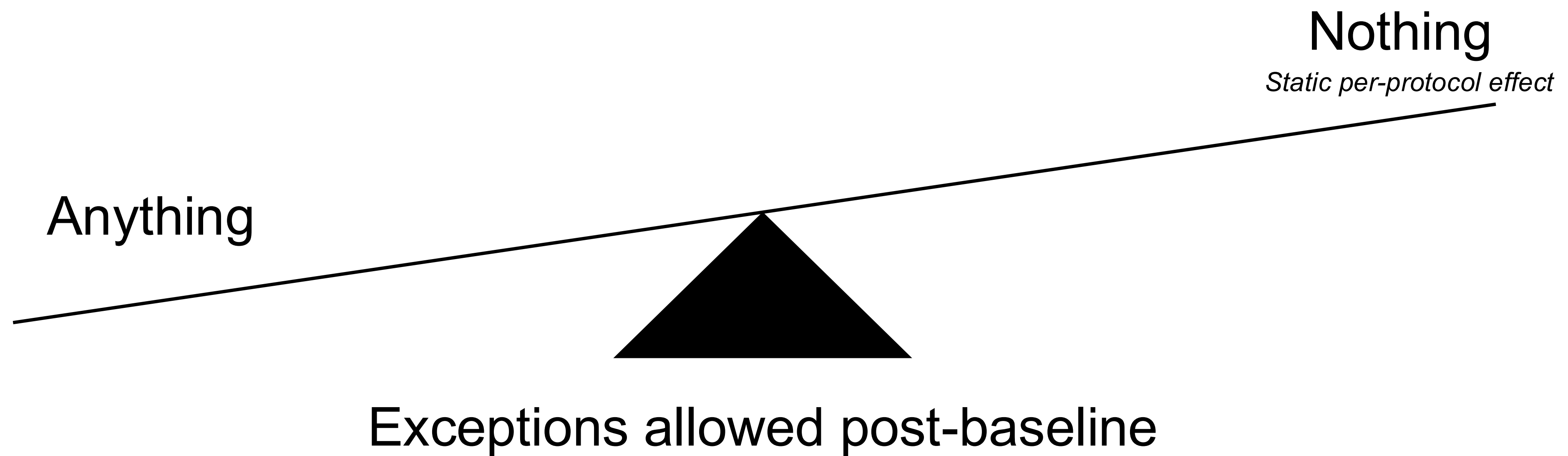


Exceptions allowed post-baseline

The trade-off



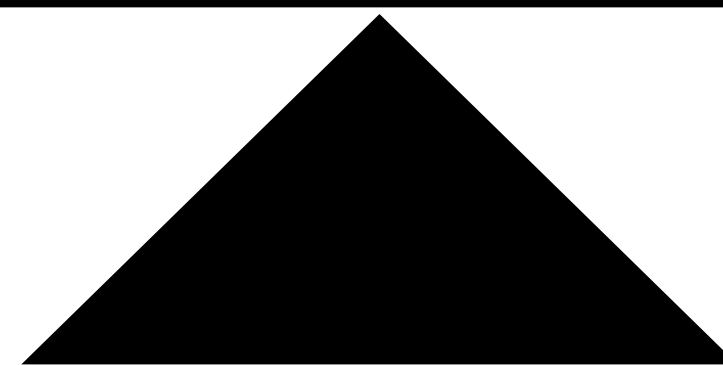
The trade-off



The trade-off

Anything

Nothing



Exceptions allowed post-baseline

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Thank you!

Questions?

Recommended Additional Reading

On the utility of the target trial framework

Hernán, M. A., Dahabreh, I. J., Dickerman, B. A., & Swanson, S. A. (2025). The target trial framework for causal inference from observational data: why and when is it helpful?. *Annals of internal medicine*, 178(3), 402-407.

On the definition of assignment in real-world data

Humphreys, A. B., Matthews, A. A., Young, J. C., Berglund, A., Lindahl, B., Wettermark, B., ... & Hernán, M. A. (2025). The definition of treatment assignment in observational emulations of target trials—an empirical examination in the Swedish Primary Care Cardiovascular Database. *Annals of Epidemiology*, 108, 56-62.

On the definition of per-protocol effects

Dahabreh, I. J., Ung, L., Hernán, M. A., & Chiu, Y. H. (2024). The role of assignment in defining and identifying causal effects in randomized trials. *arXiv preprint arXiv:2408.14710*.